Language Organization in Multilingual LLMs through SAE Feature Suppression

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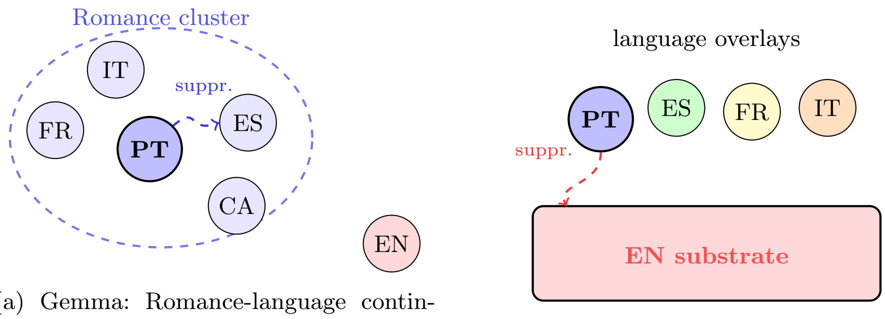
Abstract. How multilingual large language models organize the languages they speak depends on the model family; we present causal evidence for this claim. Using Portuguese as a probe, we apply Sparse Autoencoder (SAE) feature suppression to Gemma-2-2B and Llama-3.1-8B: ablating the top Portuguese-specific features redirects Gemma to Spanish or Catalan and Llama to English. These fallback directions expose internal geometries we name the *Romance-language continuum* (languages arranged by typological proximity, in Gemma) and the *English substrate* (non-English languages as overlays on an English default, in Llama). Portuguese surfaces this contrast because its Romance neighbors make any Romance fallback observable; a typologically isolated probe could not separate the two. The contrast survives feature-agnostic controls: Gemma's Romance fallback requires PT-targeted suppression, while Llama collapses to English under PT-targeted, EN-targeted, and L0-matched random suppression alike (Mann-Whitney $p = 0.88$ between PT and random), evidence that the substrate manifests as a generalized fragility. Probes in Spanish, Italian, and French confirm that Llama collapses to English on 42 of 45 Romance probes while Gemma preserves the source language. Masked FLORES cosine similarity (PT-ES > PT-EN at $p < 10^{-150}$ per layer in both models) and terminal-layer linear CKA (PT-EN at 0.61 in Gemma vs. 0.74 in Llama) corroborate the divide. Training 16 custom SAEs, we find that the ReLU+L1 vs. TopK reconstruction gap does not transfer across models (TopK FVE 79.7% on Gemma against 98.0% on Llama). To our knowledge, this is the first mechanistic study establishing a causal cross-family contrast of language-organization paradigms.

Keywords: Mechanistic Interpretability · Sparse Autoencoders · Multilingual LLMs · Portuguese NLP · Feature Steering

1 Introduction

Modern large language models (LLMs) process dozens of languages simultaneously, yet how they organize those languages internally is not well understood.



(a) Gemma: Romance-language continuum

(b) Llama: English substrate

Fig. 1: Two architectures of multilingualism made visible through SAE feature-suppression. **(a)** In Gemma-2-2B, Portuguese sits inside a Romance cluster; suppressing PT-specific features redirects generation to the nearest typological neighbor (Spanish or Catalan). **(b)** In Llama-3.1-8B, non-English languages operate as overlays on an English substrate; suppressing PT-specific features strips the Portuguese overlay and reveals the English default.

The literature offers two partial answers that have not been reconciled. Wendler et al. [20] argue that Llama models use English as an internal *lingua franca*, with non-English inputs traversing English-centric intermediate representations. A separate line of work [14] documents shared multilingual spaces where languages cluster by typological similarity. Both pictures are backed by evidence, but the evidence comes from different models, methods, and analyses, leaving open whether multilingual LLMs share one organizational paradigm or several.

This paper offers causal evidence that both pictures are correct, each for a different model family. Sparse autoencoders (SAEs) decompose a model's internal activations into a large dictionary of sparse, mostly monosemantic features that are more interpretable than raw neurons and can be edited individually, which lets us treat a language as a set of features and switch it off. Using Portuguese as a probe in SAE feature-suppression experiments (Figure 1), we find that suppressing the top Portuguese-specific features redirects generation to Spanish or Catalan in Gemma-2-2B and to English in Llama-3.1-8B. The two fallback directions expose internal geometries we name the *Romance-language continuum*, in which languages are organized by typological proximity (observed in Gemma), and the *English substrate*, in which non-English languages behave as overlays on an English default (observed in Llama).

The choice of Portuguese is methodological rather than incidental. Its Romance neighbors (Spanish, Catalan, French, Italian) make any Romance fallback directly visible, so a continuation in Spanish or Catalan after suppression in Gemma reads off the model's next-nearest language in representational space; a typologically isolated probe with no close neighbors would conflate the continuum with the substrate. Portuguese has over 260 million speakers, with Brazilian

Portuguese the dominant variety, and it remains under-represented in mechanistic interpretability work.

We make three contributions. First, we show that two distinct organizations of multilinguality coexist in current LLMs (the Romance-language continuum and the English substrate), and establish the contrast through causal feature suppression rather than correlational analysis. Second, the direction of generation collapse under suppression is itself a *lesion-based readout* of a language's representational neighborhood, and pairing it with a typologically embedded probe makes the paradigms distinguishable; the contrast survives an L0-matched random-feature control and generalizes from Portuguese to Spanish, Italian, and French probes (Llama collapses to English on 42 of 45 Romance probes). Third, three complementary measurements corroborate the divide: pairwise masked FLORES cosine similarity, linear CKA at the terminal layer (PT-EN gap of 0.13 between Gemma and Llama), and single-feature steering that surfaces a PT-EN bridge feature in Llama with no Gemma analogue.

2 Related Work

Sparse autoencoders decompose activations into overcomplete, sparse features that are more interpretable than raw neurons [7,2]. Gao et al. [10] scale SAE training and introduce the FVE and dead-feature metrics we adopt, and Shu et al. [18] survey the area. We build on pretrained SAE suites for Gemma [15] and Llama [13] and extend feature-clamping [2] to language identity.

The multilingual-representation literature is split. One line argues that Llama-family models reason through English-centric intermediates: Wendler et al. [20] use logit-lens analyses on Llama 2, and Schut et al. [17] extend the picture with activation steering, reporting that English-derived steering vectors transfer better than those from other languages. A separate line [14] documents shared multilingual spaces where languages cluster by typological similarity, and the closest cross-family concurrent work (Brinkmann et al. [3], who train SAEs on Llama-3-8B and Aya-23-8B) shows that morphosyntactic concepts such as number, gender, and tense live on feature directions shared across typologically diverse languages. We read these pictures as compatible once paired with a model family: grammatical abstraction may be shared cross-lingually while language-identity geometry remains family-dependent. Our contribution makes that divide causal and family-contrastive: suppression redirects Gemma to a Romance neighbor and Llama to English, with Portuguese serving as the typologically embedded probe that makes the two paradigms distinguishable. Concurrent SAE-steering studies target typologically distant pairs within a single model (EN→ZH, EN→JA): SASFT [8] and Causal Language Control [5] (90% single-feature steering in Gemma-2B); our setting differs by being cross-family and Romance-focused, where shared representations make steering harder. Closest in method, recent SAE work already identifies and ablates language-specific features within a single model: Deng et al. [9] show that ablating a language's SAE features degrades that language while leaving others intact and use them

Table 1: Base models and pretrained SAEs used in this study.

Model	Params	Layers	d_{model}	SAE d_{sae}	SAE Activation
Gemma-2-2B	2B	26	2,304	16,384	ReLU
Llama-3.1-8B	8B	32	4,096	32,768	JumpReLU

to control the output language, and SAE-LAPE [1] localizes language-specific features by activation probability, mostly in middle-to-final layers. Our contribution is therefore not feature suppression in isolation but the cross-family contrast it exposes: the same intervention reveals a Romance continuum in one family and an English substrate in the other. On Portuguese itself, prior NLP work targets downstream performance [19]; we provide the first mechanistic catalog of Portuguese-specific features with causal validation.

3 Methodology

Models and data. We analyze two model families (Table 1), chosen for their architectural differences and availability of pretrained SAEs.

For multilingual evaluation we use FLORES-200 [6] parallel sentences in five languages (PT, EN, ES, FR, IT), 1,012 pairs per language; because the sentences are parallel translations, differential activation patterns reflect linguistic rather than semantic differences. SAE training data comes from Portuguese Wikipedia ($\sim$2M tokens per model), tokenized with each model's native tokenizer (SentencePiece for Llama, BPE for Gemma); residual stream activations are extracted with TransformerLens [16] and stored as float16 tensors. We also analyze Gemma-2-9B (42 layers, d_{model}=3,584) for a within-family cross-scale comparison, using raw neuron activations since Gemma Scope SAEs were not available for the 9B model at the time of this study.

We compare two SAE architectures, both trained on layer-10 residual stream activations from Portuguese Wikipedia.

ReLU+L1. The encoder maps an input activation $\mathbf{x} \in \mathbb{R}^d$ to a sparse code $\mathbf{f} = \text{ReLU}(W_{\text{enc}}(\mathbf{x} - \mathbf{b}_{\text{pre}}) + \mathbf{b}_{\text{enc}}) \in \mathbb{R}^{d_{\text{sae}}}$, which the decoder reconstructs as $\hat{\mathbf{x}} = W_{\text{dec}}\mathbf{f}$. Training minimizes:

$$\mathcal{L}_{\text{ReLU}} = \|\mathbf{x} - \hat{\mathbf{x}}\|_2^2 + \lambda\|\mathbf{f}\|_1 \tag{1}$$

with decoder columns normalized to unit norm at each step.

TopK. Same encoder-decoder architecture, but enforces exact sparsity by retaining only the top-k encoder activations (applying ReLU to them) and zeroing the rest. The loss is MSE only:

$$\mathcal{L}_{\text{TopK}} = \|\mathbf{x} - \hat{\mathbf{x}}\|_2^2 \tag{2}$$

since sparsity (the ℓ_0-norm $\|\mathbf{f}\|_0 = k$ exactly) is enforced structurally rather than via penalty. We use L_0 throughout to refer to this ℓ_0-norm sparsity count.

For each base model we train 8 SAE configurations (totaling 16 custom SAEs): both architectures at $d_{\text{sae}} \in \{4{,}096, 16{,}384\}$, with $\lambda \in \{5 \times 10^{-4}, 10^{-4}\}$ for ReLU+L1 and $k \in \{32, 64\}$ for TopK. Training uses Adam (lr 3×10^{-4}) for 1 epoch with batch size 4,096 (Gemma) or 2,048 (Llama).

Differential feature analysis. To identify Portuguese-specific features, we compute a differential activation score. For each SAE feature i, we first compute the mean activation $\bar{a}_i^l$ across all FLORES tokens in language l. The differential activation score is:

$$s_i = \frac{\bar{a}_i^{\text{PT}} - \frac{1}{|\mathcal{L}'|} \sum_{l \in \mathcal{L}'} \bar{a}_i^l}{\sigma_i} \tag{3}$$

where $\mathcal{L}' = \{\text{EN, ES, FR, IT}\}$ is the set of contrast languages, $\mathcal{L} = \mathcal{L}' \cup \{\text{PT}\}$ is the full set of five FLORES languages, and $\sigma_i = \text{std}(\{\bar{a}_i^l\}_{l \in \mathcal{L}})$ normalizes by cross-language variability. Features with $s_i > 2$ are considered strongly Portuguese-specific: their Portuguese activation is more than 2 standard deviations above the cross-language mean. Unlike activation-probability methods that flag a feature by how often it fires on a language [1], Eq. 3 contrasts mean activation magnitudes and normalizes by cross-language variability, so it rewards features that are both strong and selective for Portuguese. The features it surfaces sit mostly at middle-to-upper layers, in line with where activation-probability methods localize language-specific concepts, with an additional early-layer syntactic peak in Gemma (Table 4).

We apply this analysis at 6 layers per model: layers 0, 5, 10, 15, 20, 25 for Gemma 2B; layers 0, 5, 10, 16, 21, 26 for Llama 8B (selected at approximately equivalent relative positions).

Monolinguality metric. To compare how sharply languages separate within each model, we compute a monolinguality metric on raw residual-stream neurons (not SAE features), so that the same procedure applies to Gemma-2-9B in the cross-scale analysis (Section 4), where no pretrained SAE suite is available. For each neuron i and language L, let $a_i(L) = |\overline{x_i^{(L)}}|$ denote the absolute mean activation of neuron i on FLORES sentences of language L. The inner mean is taken over content tokens only: padding (token 0 in Gemma; the eos token in Llama) and BOS are excluded, then averaged across sentences. Masking matters because FLORES sentences tokenize to different lengths (Llama: PT 39.5 vs. EN 26.8 content tokens out of a 128-token window on average), and unmasked aggregation lets padding representations dominate the cross-language comparison. A neuron is *PT-dominant* at a given layer if $\arg\max_{L \in \mathcal{L}} a_i(L) = \text{PT}$. For each PT-dominant neuron, its monolinguality is

$$m_i(\text{PT}) = \frac{a_i(\text{PT})}{\sum_{L \in \mathcal{L}} a_i(L)}, \qquad \mathcal{L} = \{\text{PT, EN, ES, FR, IT}\}. \tag{4}$$

Table 2: Best SAE per architecture per model (evaluated on FLORES PT activations, layer 10). FVE = Fraction of Variance Explained. L_0 = mean number of non-zero features per activation (ℓ_0-norm). Dead = fraction of features never activated.

Model	Architecture	FVE (%)	L_0	MSE	Dead (%)
Gemma 2B	ReLU+L1 (d=4k, λ=1e-4)	99.19	4,040	0.063	0.0
	TopK (d=16k, k=64)	79.65	64	1.578	35.3
Llama 8B	ReLU+L1 (d=16k, λ=1e-4)	**99.83**	7,684	0.001	1.1
	TopK (d=16k, k=64)	98.01	64	0.009	31.3

Values close to $1/|\mathcal{L}| = 0.2$ indicate a neuron that activates comparably across all languages; values near 1 indicate a neuron that fires almost exclusively on Portuguese. We report, per model, the mean of $m_i(\text{PT})$ over PT-dominant neurons at each analyzed layer, and a single per-model figure obtained by averaging across those layers. For raw-neuron analyses (monolinguality and the pairwise similarity of Section 5) we use the per-model layers listed above and add Llama's terminal layer 31, so the per-model average is taken over 6 layers for Gemma and 7 for Llama, placing each model's deepest analyzed layer at 100% relative depth.

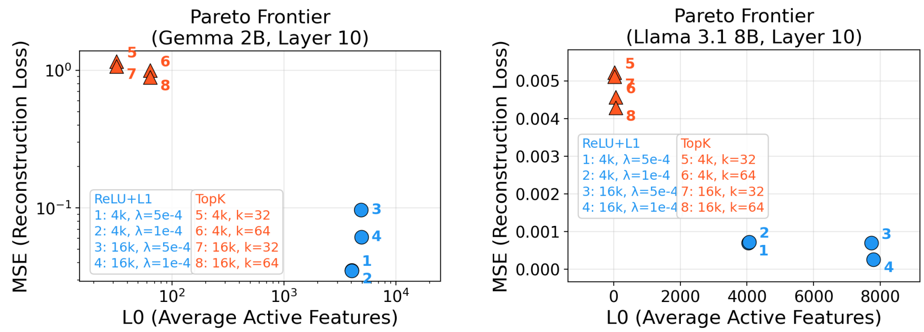
Steering experiments. We perform causal interventions by hooking into the model's forward pass at layer 10 and modifying SAE-encoded features. For amplification, we encode the residual activation through the SAE, multiply the top-20 Portuguese-specific features by factor $\alpha \in \{1, 2, 5, 10\}$, decode back, and replace the original activation; this tests whether increasing PT feature magnitudes forces Portuguese generation. For suppression, the same procedure is applied but the top-20 features are set to zero, testing whether removing PT features causes language switching and, in particular, which language the model defaults to.

For Gemma 2B we use Gemma Scope (layer 10, 16k features); for Llama 8B we use Llama-Scope (layer 10, 32k features, JumpReLU). We test on 5 prompts per language (PT, EN, ES, FR), generating 50 tokens with temperature 0.7.

4 Experiments and Results

Reconstruction quality. Table 2 reports the best configuration per architecture for each model; Figure 2 shows the full Pareto frontier across all 16 custom SAEs.

TopK performance is model-dependent: 79.7% FVE on Gemma 2B (35.3% dead) versus 98.0% on Llama 8B (31.3% dead). ReLU+L1 exceeds 99% FVE on both models with near-zero dead rates, making it the more robust choice; the Discussion (Section 5) examines the likely source of the gap.



(a) Gemma 2B (b) Llama 3.1 8B

Fig. 2: Pareto frontier: reconstruction loss (MSE) vs. sparsity (ℓ_0-norm L_0) for all 8 SAE configurations per model. Blue circles = ReLU+L1; orange triangles = TopK. Gemma axes are on a log scale; Llama axes are linear.

Table 3: Language classification accuracy (5-fold CV, 1,012 samples per language). The best custom SAE (by FVE) is used for SAE feature probing.

Representation	**Gemma 2B**	**Llama 8B**
Raw activations (d_{model})	99.94% ± 0.12	99.98% ± 0.04
SAE features (d_{sae})	99.98% ± 0.04	99.88% ± 0.15

Language probing. Linear probing (logistic regression, 5-fold CV) on mean-pooled layer-10 activations for 5-way language classification yields near-perfect accuracy for both models (Table 3), confirming that language identity is linearly separable in the residual stream and that SAE encoding preserves this information.

All accuracies exceed 99.8%, meaning a simple linear classifier can distinguish languages from a single layer's activations. SAE features achieve comparable accuracy to raw activations despite imperfect reconstruction (FVE of 99.2% for the Gemma SAE and 99.8% for the Llama SAE used for probing), confirming that the SAE decomposition retains language-relevant information.

Portuguese-specific features. Differential analysis (Eq. 3) with pretrained SAEs identifies strongly Portuguese-specific features ($s_i > 2$) across all analyzed layers in both models (Table 4).

Both models contain thousands of strongly PT-specific features. Normalizing for SAE dictionary size: Gemma has 4,482/16,384 = 27.4% and Llama has 3,811/32,768 = 11.6%. Gemma 2B concentrates features at layer 0 (1,134) and recovers a second peak at layer 25 (777); Llama 8B peaks at layer 10 (747) and declines to 452 at layer 26.

From the set of strongly PT-specific features in Gemma 2B we selected the top 20 features per layer by differential score, yielding 120 features across the

Table 4: Strongly PT-specific feature counts ($s > 2$) at each analyzed layer using pretrained SAEs (column headers indicate layer indices; Gemma 2B: layers 0, 5, 10, 15, 20, 25; Llama 8B: layers 0, 5, 10, 16, 21, 26). Both rows use padding-masked analysis: tokens identified as padding or BOS are excluded before computing per-feature mean activations. The differential activation score has a theoretical ceiling of $5/2 = 2.5$ (when a feature activates only on PT among the five languages, its score equals mean/std = 2.5).

Model (SAE)	Layer 0	Layer 5	Layer 10	Layer 15/16	Layer 20/21	Layer 25/26	Total
Gemma 2B (16k)	1,134	835	733	473	530	777	4,482
Llama 8B (32k)	698	675	747	646	593	452	3,811

Table 5: Categories of Portuguese-specific features in Gemma 2B (120 features across 6 layers, classified by their top-10 activating content tokens). Counts assigned by a deterministic rule-based classifier; the residual "Mixed" category covers features whose top tokens span more than one of the listed classes.

Category	L0	L5	L10	L15	L20	L25	Total	Example tokens
Lexical (content)	3	10	11	9	5	6	44	"Sul", "modern", "cit"
Syntactic (function)	15	3	2	2	4	5	31	"de", "do", "os", "uma"
Mixed	1	1	3	2	2	5	14	heterogeneous
Lexical (proper noun)	0	3	2	1	4	0	10	"Brasil", "Portugal"
Punctuation	0	1	1	4	2	2	10	".", ",", "("
Subword fragment	1	2	1	2	3	1	10	"rália", "-icos"
Whitespace	0	0	0	0	0	1	1	" "

6 analyzed layers. We categorize each feature by its 10 top-activating content tokens in FLORES PT using a rule-based classifier that checks for function words, PT-typical suffix patterns, capitalized proper nouns, punctuation, and whitespace (Table 5).

A layer-wise gradient emerges: layer 0 is dominated by syntactic features (15 of 20 features are function words: "de", "do", "os", "uma"), middle layers (5–15) shift toward lexical content words and proper nouns, and later layers retain a mix of syntactic, punctuation, and content features.

For Llama 8B, many top features initially activated on padding tokens, an artifact of differential tokenization lengths (PT averages 39.5 content tokens per sentence vs. EN's 26.8). Masking padding positions before computing mean activations reduces the strongly PT-specific count from 4,143 to 3,811 (-8%); Table 6 shows the corrected top-20.

For Llama we inspected the top-20 features at layer 10, against 120 across 6 layers for Gemma, reflecting the cost of token-level inspection of Llama-Scope's larger dictionary. After masking, syntactic features (prepositions "de"/"com", articles "a"/"os"/"um") dominate, with morphological features ("são", "â", suffixes "-icos") also present, mirroring Gemma's layer-10 composition. Methodologically,

Table 6: Top-20 Llama-Scope PT features (layer 10) after padding masking. Features are categorized by their top-activating tokens on genuine (non-padding) Portuguese text.

Category	**Count**	**Example tokens**	**Note**
Syntactic	14	“de”, “os”, “a”, “um”, “que”, “com”	Prepositions, articles, conjunctions
Morphological	4	“são”, “â”, “-icos”, “-ode”	PT verb forms, suffixes
Mixed/subword	2	“ab”, “di”	Subword fragments in PT contexts

differential analysis on padded sequences can produce spurious results when languages tokenize to different lengths, a pitfall we have not seen documented in the SAE interpretability literature.

Steering results. Table 7 delivers our central result: suppressing the top-20 PT-specific features redirects generation in family-specific directions, and that direction distinguishes the two paradigms of Figure 1.

In Gemma 2B, suppression consistently lands on a closely related Romance language: Spanish for most prompts, and Catalan for *“A capital do Brasil é”*. The Catalan output is informative because Catalan is not among the five FLORES evaluation languages, so the fallback is not a ranking over the measured set but reflects a Romance geometry that includes unevaluated varieties. For *“A língua portuguesa é falada em”* the Spanish continuation also preserves topical coherence: the model keeps talking about languages, just in another Romance one. Llama 8B, in contrast, falls back to English on every suppression prompt, consistent with the English-pivot picture of Wendler et al. [20]: non-English surface forms behave as overlays on an English-centric base, so stripping the Portuguese overlay reveals it.

Two controls test whether the fallback direction is driven by PT-feature targeting or by any feature-level perturbation: an L0-matched random condition (20 features matched in PT activation rate but with $|s| \leq 1$, three random seeds) and the top-20 EN-specific features. Across 45 generations per random condition and 15 per EN condition (5 prompts $\times$ 3 generation seeds), Gemma preserves Portuguese under both controls (random: 36/45; EN: 13/15) while only PT-feature suppression breaks Portuguese (11/15 shift to a Romance language). Llama collapses to English at the same rate under PT, EN, and random conditions (10/15, 10/15, 31/45 respectively; Mann-Whitney $p = 0.88$ for PT vs. random). Extending the protocol to Spanish, Italian, and French probes, where we suppress the top-20 features specific to each language on prompts in that language, Llama collapses to English on 42 of 45 Romance probes (ES 14/15, IT 15/15, FR 13/15) while Gemma preserves the source language (ES 12/15, IT 15/15, FR 15/15). Gemma's Romance-language collapse is therefore feature-specific, while Llama's English collapse reflects a generalized fragility of language identity at layer 10 that generalizes to every Romance overlay we tested.

Amplification exposes a complementary asymmetry. Multiplying the top-20 PT features by 10 leaves 18 of 25 Gemma generations (5 prompts $\times$ 5 seeds) in

Table 7: Steering examples. Baseline: normal generation; Suppression: top-20 PT features set to zero; Amplification: top-20 PT features multiplied by 10. Fallback-language labels (**in bold**) were assigned by manual inspection of the generated text.

Prompt	Mode	Gemma 2B output	Llama 8B output
A capital do Brasil é	Base	palco de diversos eventos e festivais...	o centro mais importante do país...
	Suppr.	la una metròpolis ambduna de les fa uns centenars... **(Catalan)**	a capital, 1 capital, is 1 capital... **(English)**
	Amplif.	uma das métroas mais importantes da metropode... **(PT, degraded)**	que que que u u u... **(collapsed)**
O rio mais longo do mundo é	Suppr.	el férriable del continente... **(Spanish)**	the longest the most long the... **(English)**
	Amplif.	o Brasil, e sua extensão é de 235 mil quilômetros... **(PT)**	mais do mais mais... **(collapsed)**
A língua portuguesa é falada em	Base	22 países, sendo Portugal um dos principais...	mais de 70 países, cerca de 200 milhões de falantes...
	Suppr.	todos os continentes, enrapapeando la lengua española... **(Spanish)**	il...?...?...? **(degenerate)**

Portuguese against only 6 of 25 in Llama, with 14 of 25 Llama outputs collapsing to English. Continuation length drops in both models (Gemma 24% [bootstrap 95% CI 11–37], Llama 16% [0–31] at factor 10), and the Mann-Whitney test does not separate the two ($p = 0.69$); the family contrast is in the language identity of the surviving output, not in the magnitude of the collapse. The most likely mechanistic explanation is sparsity: Llama-Scope features fire on roughly $L_0 = 24$ feature dimensions per content token vs. $L_0 = 86$ for Gemma Scope, so 20-feature interventions perturb 83% of the active code in Llama against 24% in Gemma. Direct magnitude comparison between the two SAEs is confounded by different activation functions (ReLU+L1 vs. JumpReLU) and input normalization, so we report sparsity as the architecture-independent quantity that supports this account.

Amplifying individual Llama-Scope features (factor 5, prompt *"A capital do Brasil é"*) exposes distinguishable specializations. Feature 16811 produces repetitions drawn exclusively from {Portugal, Portuguese, Brazil}, a "Lusophone-world" concept; feature 18397 generates a PT-EN hybrid (*"a capital do Brazil is the capital of Britain countries"*), a cross-lingual bridge that surfaces the substrate at the feature level; feature 16096 yields Brazil-anchored repetitions (*"BC Brazil is BC Brazil is is is..."*). The bridge feature has no direct Gemma ana-

Table 8: Cross-scale comparison within the Gemma family. Cosine similarity between masked-mean PT and ES/EN/FR activations and the count of strongly PT-specific neurons ($s > 2$, applied to raw neurons rather than SAE features). All entries use the same padding-masked aggregation as the monolinguality metric of Eq. 4.

9B Layer	**Rel. pos.**	**cos(PT,ES)**	**cos(PT,FR)**	**cos(PT,EN)**	$n_{s>2}$
0	0%	0.959	0.944	0.887	234
8	20%	0.982	0.977	0.957	302
16	39%	0.996	0.995	0.988	183
24	59%	0.994	0.991	0.986	252
32	78%	0.982	0.960	0.962	232
41	100%	0.640	0.519	0.461	402

logue in our inspected inventory, matching a picture in which Gemma's Romance languages share representational space directly rather than routing through English.

Progressive steering on Gemma 2B localizes where language identity lives. Amplifying the top-20 PT-specific features (factor 5) at an increasing set of layers and scoring continuations with a PT-likeness heuristic $0.3\,c + 10\,w$ (Portuguese-specific characters c and function-word ratio w), layer 25 alone yields 4.5, layers 15+20+25 together peak at 7.2, and adding layer 10 drops the score to 0.9 (incoherent text). The heuristic agrees with fastText LID on the coherent subset (Spearman $\rho = 0.83$, $n = 55$); published LIDs collapse on the degenerate outputs of this experiment, so marker counting is the appropriate readout. Language identity therefore lives in mid-to-late layers (15–25); steering early layers disrupts foundational representations that later layers depend on, which is why multi-layer suppression succeeds where single-layer steering does not.

Cross-scale analysis. To disentangle family effects from scale effects, we also analyzed Gemma-2-9B (42 layers, $d_{\text{model}} = 3{,}584$) using raw neuron activations (Table 8).

The 9B model shows progressive linguistic differentiation (Figure 3): cosine similarity between Portuguese and Spanish masked-mean activations decreases from 0.959 at layer 0 to 0.640 at layer 41, with the sharpest drop in the final third. The PT-EN drop is steeper still (0.887 $\rightarrow$ 0.461) and a clean Iberian ordering emerges at the terminal layer (ES > FR > EN: 0.640, 0.519, 0.461). The number of strongly PT-specific neurons rises from 234 at layer 0 to 402 at layer 41. The 2B model, by contrast, shows a flatter trajectory at the analyzed layers under the same masked aggregation. We rely on the qualitative contrast (flat vs progressive) rather than absolute magnitudes, since the 2B and 9B numbers live in different spaces (2B uses Gemma Scope SAE features, 9B uses raw neurons, as no pretrained SAE suite was available for the 9B model). The pattern suggests that larger models develop finer language separation in deeper layers, a property not visible at the 2B scale.

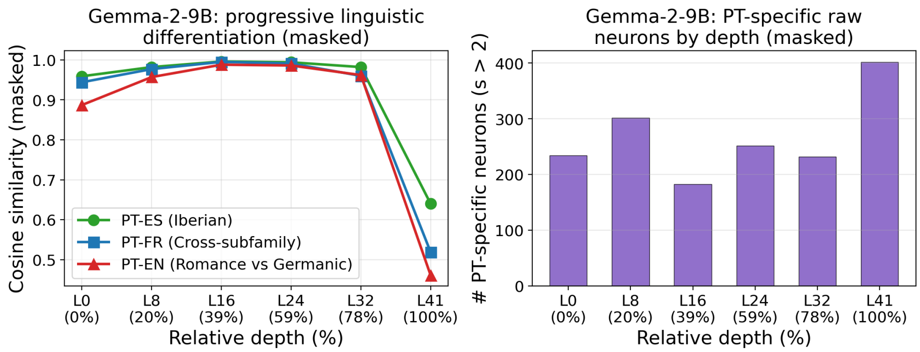


Fig. 3: Progressive linguistic differentiation in Gemma-2-9B: masked cosine similarity between Portuguese and Spanish, French, and English mean activations decreases in deeper layers (left), while the number of PT-specific neurons increases (right). This gradient is absent in the 2B model.

5 Discussion

Cross-family language organization. A pairwise cosine similarity analysis on padding-masked FLORES activations supports a layered typological separation in both models: PT-EN distance (Romance vs. Germanic) is already the largest at the earliest layers, PT-FR distance (cross-subfamily) grows through middle layers, and PT-ES distance (Iberian) diverges only at the deepest analyzed layer. The PT-ES vs. PT-EN gap is reproducible at every layer in both models when computed per sentence (paired Wilcoxon, $p < 10^{-150}$ at every layer with $N = 1012$ sentences; $p < 10^{-149}$ after Bonferroni correction across the 13 analyzed layers). Linear CKA between PT and other-language sentence representations at the terminal layer reproduces this ordering and quantifies the cross-model contrast: PT-ES CKA is 0.80 in both Gemma and Llama, while PT-EN CKA is 0.61 in Gemma against 0.74 in Llama, a 0.13 gap that is the single strongest quantitative signature of the English-substrate paradigm in our data.

The texture of the fallback strengthens the picture. Gemma drifts into Catalan on *"A capital do Brasil é"*, a language absent from our five evaluation languages, indicating a continuum that reaches beyond the measured set; Gemma 2's training distribution [11] plausibly encourages this shared Romance structure, although Google does not publicly detail its language composition. Llama falls back to English consistent with Wendler et al. [20] and with Meta's predominantly English pretraining mix for Llama 3 (only 8% multilingual tokens) [12], which would make English a natural base layer.

The padding-masked monolinguality metric of Eq. 4 does not by itself separate the two models: averaged across analyzed layers, Llama has PT monolinguality 0.367 (95% CI 0.348–0.386) against Gemma's 0.363 (0.336–0.391), a paired difference of 0.004 over relative-depth-matched layers (Wilcoxon paired

$p = 0.28$). The substrate paradigm is supported instead by the suppression and probe experiments (Section 4) and by the CKA gap reported above. Among individual features, amplifying Llama-Scope feature 18397 produces a PT-EN hybrid (*"a capital do Brazil is the capital of Britain countries"*) with no analogue in our inspected Gemma inventory, consistent with a Llama representation in which Romance overlays route through English rather than sharing space with one another. These suppression experiments may also be read as intentionally induced code-switching, connecting to SASFT [8]. One mechanistic reading of the English fallback is that Llama's task and instruction processing is anchored in English: Cho and Hockenmaier [4] identify task-instruction-focused SAE features in multilingual models, and if such features are English-centric in Llama, perturbing any non-English overlay would expose the English default beneath rather than a Romance neighbor. This is consistent with the collapse being insensitive to which features are removed: PT-targeted and L_0-matched random suppression both drive Llama to English (Mann-Whitney $p = 0.88$), so the substrate reads as a generalized fragility of non-English identity rather than a property of the specific Portuguese features.

The two paradigms suggest different affordances: a Romance continuum may permit finer-grained steering between related languages, while an English substrate raises concerns about English-centric biases propagating to other languages.

TopK performance across models. The TopK gap (79.7% FVE on Gemma 2B vs. 98.0% on Llama 8B) appears under identical hyperparameters. A plausible explanation is intrinsic dimensionality: Llama's d_{model}=4,096 (nearly twice Gemma's 2,304) lets activations concentrate on a k-sparse basis more readily, while Gemma's lower-dimensional activations diffuse across many components and defeat exact-k sparsity. ReLU+L1 adapts its sparsity per sample and exceeds 99% FVE on both models; TopK produces 31% dead features on Llama and 35% on Gemma at the best configuration, versus <1.2% for ReLU+L1. The practical takeaway is that TopK transfer is not automatic: practitioners should validate on their specific base model.

Limitations. Our two models differ simultaneously in family, size, attention variant, and training mix, so the observed differences may reflect any combination of these factors. Scale alone is unlikely to explain the paradigm: Gemma-2-9B (larger than Llama-3.1-8B) preserves a Romance-continuum organization through its terminal layer (cos(PT,ES)=0.640 vs. cos(PT,EN)=0.461, Table 8) rather than drifting toward an English-centric geometry. A fully controlled comparison would still require same-size models from different families. The pretrained SAEs differ in activation function, dictionary size, and training procedure, so we rely on qualitative contrasts and on the architecture-independent sparsity statistic (L_0). We report only padding-masked values; an earlier unmasked draft overstated the Llama vs. Gemma monolinguality gap and understated the typological-continuum signal in pairwise cosines. Language identification uses fastText LID-176 top-1 with no confidence threshold, and bootstrap confidence intervals use a fixed set of five Portuguese prompts; generalization beyond this set is not directly tested. Steering generation uses T=0.7, so the specific strings in Table 7 are illustrative.

In scope, we analyze 6 of 26–32 layers per model, 5 European languages (4 Romance), and a single corpus (FLORES-200); the European focus may inflate Portuguese specificity scores and other genres may activate different features. Future work should extend to additional model families (Mistral, Qwen) and to same-size comparisons, include typologically diverse languages, and investigate PT-BR vs. PT-PT dialect variation.

6 Conclusion

We presented causal evidence for two architectures of multilingualism. Suppressing Portuguese-specific features redirects Gemma-2-2B to Spanish or Catalan and Llama-3.1-8B to English, exposing internal geometries we name the Romance-language continuum and the English substrate. The contrast survives an L0-matched random-feature control (Gemma's Romance fallback requires PT-targeted suppression; Llama collapses to English under any matched 20-feature perturbation) and generalizes to Spanish, Italian, and French probes (Llama 42 of 45 to English; Gemma preserves the source language). Pairwise masked FLORES cosine similarity, linear CKA at the terminal layer (PT-EN gap of 0.13 between Gemma and Llama), and single-feature steering that surfaces a PT-EN bridge feature in Llama with no Gemma analogue corroborate the divide. Portuguese is what makes the paradigms distinguishable: a typologically embedded probe reveals the continuum where an isolated probe could not. Contrasting two families rather than describing one suggests that training data composition shapes the organizational paradigm itself, a hypothesis further cross-family studies can test.

Reproducibility. Source code, configurations, per-experiment result files, and the figures reported here are available at https://github.com/joaopaulopresa/llm-pt-interpretability[4].

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